

Thu, Jul 2 at 4:18 PM

I need Feynman diagrams for calculating the self-energy of Hubbard model on 3D square lattice up to order 4 in U , and the exact expressions over k -momenta and/or energies that I need to evaluate them.

I will assume a simple-cubic Hubbard model with nearest-neighbor hopping and use bare Matsubara perturbation theory for $Un_{\uparrow}n_{\downarrow}$. I'll make momentum/frequency normalization, spin factors, and bare versus skeleton conventions explicit.

Thought for 7m 44s

I assume the **normal, paramagnetic, single-band Hubbard model** on a **3D simple-cubic lattice**. The diagram algebra is dimension-independent; the 3D lattice enters only through the dispersion and the Brillouin-zone integral. The diagram labels below follow the standard fourth-order Hubbard self-energy notation (2), (3RPA), (3VC), (4A) . . . (4L), whose line drawings and analytic expressions are given explicitly in Appendix A of Shinkai–Ikeda–Yamada for the 2D case; replacing the dispersion and BZ gives the 3D simple-cubic result. [arXiv +1](#)

1. Model and notation

Use lattice constant $a = 1$, first Brillouin zone $[-\pi, \pi)^3$, and

$$H = \sum_{\mathbf{k}\sigma} \xi_{\mathbf{k}} c_{\mathbf{k}\sigma}^{\dagger} c_{\mathbf{k}\sigma} + \frac{U}{N} \sum_{\mathbf{k}\mathbf{p}\mathbf{q}} c_{\mathbf{k}+\mathbf{q},\uparrow}^{\dagger} c_{\mathbf{k},\uparrow} c_{\mathbf{p}-\mathbf{q},\downarrow}^{\dagger} c_{\mathbf{p},\downarrow},$$

with nearest-neighbor simple-cubic dispersion

$$\varepsilon_{\mathbf{k}} = -2t (\cos k_x + \cos k_y + \cos k_z), \quad \xi_{\mathbf{k}} = \varepsilon_{\mathbf{k}} - \mu_0.$$

The bare Green function is

$$G_0(k) = \frac{1}{i\omega_n - \xi_{\mathbf{k}}}, \quad k \equiv (\mathbf{k}, i\omega_n), \quad \omega_n = (2n + 1)\pi T.$$

For bosonic four-momenta,

$$q \equiv (\mathbf{q}, i\Omega_m), \quad \Omega_m = 2m\pi T.$$

I use

$$\int_k^F \equiv T \sum_{\omega_n} \int_{\text{BZ}} \frac{d^3 k}{(2\pi)^3}, \quad \int_q^B \equiv T \sum_{\Omega_m} \int_{\text{BZ}} \frac{d^3 q}{(2\pi)^3}.$$

On a finite $N = L^3$ lattice, replace

$$\int_{\text{BZ}} \frac{d^3 k}{(2\pi)^3} \longrightarrow \frac{1}{N} \sum_{\mathbf{k}}.$$

The first-order Hartree term is

$$\Sigma_{\sigma}^{(1)} = U n_{-\sigma}^{(0)}, \quad n_{\sigma}^{(0)} = \int_{\text{BZ}} \frac{d^3 k}{(2\pi)^3} f(\xi_{\mathbf{k}}).$$

For a paramagnet,

$$\Sigma^{(1)} = U \frac{n^{(0)}}{2}.$$

Usually this is absorbed into the chemical potential. At half filling on the particle-hole symmetric simple-cubic lattice, one often uses $\mu = U/2$, or equivalently a Hartree-subtracted interaction, so the nontrivial dynamical self-energy starts at U^2 .

2. Bubbles used in the diagrams

Define the particle-hole and particle-particle bubbles

$$\chi_0(q) = - \int_k^F G_0(k+q) G_0(k),$$

$$\phi_0(q) = - \int_k^F G_0(q-k) G_0(k).$$

After performing the internal fermionic Matsubara sum,

$$\chi_0(\mathbf{q}, i\Omega_m) = \int_{\text{BZ}} \frac{d^3 p}{(2\pi)^3} \frac{f(\xi_{\mathbf{p}+\mathbf{q}}) - f(\xi_{\mathbf{p}})}{i\Omega_m + \xi_{\mathbf{p}} - \xi_{\mathbf{p}+\mathbf{q}}},$$

and

$$\phi_0(\mathbf{q}, i\Omega_m) = \int_{\text{BZ}} \frac{d^3 p}{(2\pi)^3} \frac{1 - f(\xi_{\mathbf{p}}) - f(\xi_{\mathbf{q}-\mathbf{p}})}{i\Omega_m - \xi_{\mathbf{p}} - \xi_{\mathbf{q}-\mathbf{p}}}.$$

The limiting forms at zero denominator are understood by continuity, e.g. the $q \rightarrow 0, \Omega \rightarrow 0$ limit of χ_0 gives $-\int_{\mathbf{p}} f'(\xi_{\mathbf{p}})$.

3. Expansion convention

Write the correlation part as

$$\Sigma_{\text{corr}}(k) = U^2 \sigma_2(k) + U^3 \sigma_3(k) + U^4 \sigma_4(k) + O(U^5).$$

This avoids an order-counting ambiguity in diagrams (4D)–(4F): below, σ_2 is the U^2 -stripped coefficient, not the full $U^2 \sigma_2$.

The complete normal self-energy through fourth order, excluding the trivial Hartree shift, is the diagram set (2), (3RPA), (3VC), (4A) . . . (4L). The source figure uses solid lines for G_0 and dashed lines for U . 

4. Second order: one diagram

Diagram (2): second-order sunset / one particle-hole bubble

$$\sigma_2(k) = \int_q^B \chi_0(q) G_0(k - q).$$

Hence

$$\Sigma^{(2)}(k) = U^2 \int_q^B \chi_0(q) G_0(k - q).$$

An explicit energy-denominator form after all internal frequency sums is

$$\Sigma^{(2)}(\mathbf{k}, z) = U^2 \int_{\mathbf{p}} \int_{\mathbf{q}} \frac{f_{\mathbf{p}} (1 - f_{\mathbf{p}+\mathbf{q}}) (1 - f_{\mathbf{k}-\mathbf{q}}) + (1 - f_{\mathbf{p}}) f_{\mathbf{p}+\mathbf{q}} f_{\mathbf{k}-\mathbf{q}}}{z + \xi_{\mathbf{p}} - \xi_{\mathbf{p}+\mathbf{q}} - \xi_{\mathbf{k}-\mathbf{q}}},$$

where $z = i\omega_n$ on the Matsubara axis or $z = \omega + i0^+$ for the retarded self-energy, and $f_{\mathbf{k}} \equiv f(\xi_{\mathbf{k}})$.

5. Third order: two diagrams

Diagram (3RPA): particle-hole RPA chain

$$\sigma_{3\text{RPA}}(k) = \int_q^B \chi_0(q)^2 G_0(k - q).$$

Diagram (3VC): vertex-correction / particle-particle channel

$$\sigma_{3\text{VC}}(k) = \int_q^B \phi_0(q)^2 G_0(q - k).$$

Thus

$$\Sigma^{(3)}(k) = U^3 [\sigma_{3\text{RPA}}(k) + \sigma_{3\text{VC}}(k)].$$

At exact particle-hole symmetry on a bipartite nearest-neighbor lattice, the non-Hartree odd-order correlation terms cancel in many standard conventions; away from that symmetry both terms should be kept.

6. Fourth order: twelve diagrams

Write

$$\sigma_4(k) = \sum_{\alpha=A}^L \sigma_{4\alpha}(k), \quad \Sigma^{(4)}(k) = U^4 \sigma_4(k).$$

(4A) and (4B): fourth-order particle-hole RPA diagrams

$$\sigma_{4A}(k) = \int_q^B \chi_0(q)^3 G_0(k - q),$$

$$\sigma_{4B}(k) = \int_q^B \chi_0(q)^3 G_0(k - q).$$

They are algebraically identical in the paramagnetic Hubbard case, so in code you can use

$$\sigma_{4A}(k) + \sigma_{4B}(k) = 2 \int_q^B \chi_0(q)^3 G_0(k - q).$$

(4C): fourth-order particle-particle ladder

$$\sigma_{4C}(k) = \int_q^B \phi_0(q)^3 G_0(q - k).$$

(4D), (4E), (4F): non-skeleton self-energy-insertion diagrams

$$\sigma_{4D}(k) = \int_p^F \chi_0(k - p) G_0(p)^2 \sigma_2(p),$$

$$\sigma_{4E}(k) = \int_p^F \chi_0(p - k) G_0(p)^2 \sigma_2(p),$$

$$\sigma_{4F}(k) = \int_p^F \phi_0(k + p) G_0(p)^2 \sigma_2(p).$$

These three are present in the **bare** G_0 expansion. In a bold-line skeleton expansion, omit them and use dressed G in the skeleton diagrams instead.

(4G): two particle-hole bubbles connected by three fermion lines

$$\sigma_{4G}(k) = \int_{q_1}^B \int_{q_2}^B \chi_0(q_1) \chi_0(q_2) G_0(k - q_1) G_0(k - q_2) G_0(k - q_1 - q_2).$$

(4H) and (4I): mixed particle-hole / particle-particle vertex diagrams

$$\sigma_{4H}(k) = - \int_{q_1}^B \int_{q_2}^B \chi_0(q_1) \phi_0(q_2) G_0(k - q_1) G_0(q_2 - k) G_0(q_1 + q_2 - k),$$

$$\sigma_{4I}(k) = - \int_{q_1}^B \int_{q_2}^B \chi_0(q_1) \phi_0(q_2) G_0(k - q_1) G_0(q_2 - k) G_0(q_1 + q_2 - k).$$

Again, for the paramagnetic case they are algebraically identical:

$$\sigma_{4H}(k) + \sigma_{4I}(k) = -2 \int_{q_1}^B \int_{q_2}^B \chi_0(q_1) \phi_0(q_2) G_0(k - q_1) G_0(q_2 - k) G_0(q_1 + q_2 - k).$$

(4J): crossed / envelope diagram with χ_0

$$\sigma_{4J}(k) = \int_q^B \int_{p_1}^F \int_{p_2}^F \chi_0(q - p_1 - p_2) G_0(p_1) G_0(p_2) G_0(q - p_1) G_0(q - p_2) G_0(q - k).$$

(4K): crossed / envelope diagram with ϕ_0

$$\sigma_{4K}(k) = \int_q^B \int_{p_1}^F \int_{p_2}^F \phi_0(q + p_1 + p_2) G_0(p_1) G_0(p_2) G_0(q + p_1) G_0(q + p_2) G_0(k - q).$$

(4L): crossed particle-hole diagram

$$\sigma_{4L}(k) = - \int_q^B \int_{p_1}^F \int_{p_2}^F \chi_0(p_1 - p_2) G_0(p_1) G_0(p_2) G_0(q + p_1) G_0(q + p_2) G_0(k - q).$$

The (4J), (4K), and (4L) diagrams are the most expensive numerically because their sums are not reducible to simple convolution forms; the source paper explicitly notes this and uses an additional real-space cutoff approximation for them, but the expressions above are the exact Matsubara sums. 

7. Compact final expression

With the Hartree term displayed separately,

$$\Sigma(k) = U n_{-\sigma}^{(0)} + U^2 \sigma_2(k) + U^3 [\sigma_{3\text{RPA}}(k) + \sigma_{3\text{VC}}(k)] + U^4 \sum_{\alpha=A}^L \sigma_{4\alpha}(k) + O(U^5)$$

or, if the Hartree shift is absorbed into μ ,

$$\Sigma_{\text{dyn}}(k) = U^2 \sigma_2(k) + U^3 [\sigma_{3\text{RPA}}(k) + \sigma_{3\text{VC}}(k)] + U^4 \sum_{\alpha=A}^L \sigma_{4\alpha}(k) + O(U^5).$$

8. Practical evaluation notes

All momenta in arguments such as $k - q$, $q - k$, $q + p_1 + p_2$ are understood modulo reciprocal lattice vectors, with fermionic or bosonic Matsubara parity fixed by the full four-momentum combination.

For numerical work, precompute G_0 , χ_0 , and ϕ_0 on the 3D momentum grid and Matsubara grid. The diagrams (2), (3), (4A)–(4F) are mostly convolution-like. (4G)–(4I) require double bosonic sums for each external k . (4J)–(4L) are the true bottleneck and often dominate the cost in 3D.

 Sources

What about real-frequency axis?

I'll recast the Matsubara expressions into retarded real-axis form, using explicit energy denominators at low order and spectral-convolution representations for higher-order diagrams.

This keeps the same diagram topology and replaces Matsubara sums directly, so no Padé continuation is needed.

Thought for 10m 49s

Yes. On the real-frequency axis you should **not** simply replace every internal Matsubara Green function by a retarded one. The correct procedure is:

$$i\omega_n \rightarrow \omega + i0^+$$

only after the internal Matsubara sums have been done, or equivalently by using spectral representations / Keldysh rules.

Below I write the directly evaluable real-axis form for the main diagram classes.

1. Real-axis bubbles

Keep the same bare dispersion

$$\xi_{\mathbf{k}} = -2t(\cos k_x + \cos k_y + \cos k_z) - \mu_0.$$

The retarded particle-hole bubble is

$$\chi_0^R(\mathbf{q}, \Omega) = \int_{\text{BZ}} \frac{d^3 p}{(2\pi)^3} \frac{f(\xi_{\mathbf{p}+\mathbf{q}}) - f(\xi_{\mathbf{p}})}{\Omega + i0^+ + \xi_{\mathbf{p}} - \xi_{\mathbf{p}+\mathbf{q}}}.$$

Its imaginary part is

$$\chi_0''(\mathbf{q}, \Omega) = \text{Im} \chi_0^R(\mathbf{q}, \Omega) = -\pi \int_{\text{BZ}} \frac{d^3 p}{(2\pi)^3} [f_{\mathbf{p}+\mathbf{q}} - f_{\mathbf{p}}] \delta(\Omega + \xi_{\mathbf{p}} - \xi_{\mathbf{p}+\mathbf{q}}).$$

Equivalently,

$$\chi_0''(\mathbf{q}, \Omega) = \pi \int_{\text{BZ}} \frac{d^3 p}{(2\pi)^3} [f_{\mathbf{p}} - f_{\mathbf{p}+\mathbf{q}}] \delta(\Omega + \xi_{\mathbf{p}} - \xi_{\mathbf{p}+\mathbf{q}}).$$

The retarded particle-particle bubble is

$$\phi_0^R(\mathbf{q}, \Omega) = \int_{\text{BZ}} \frac{d^3 p}{(2\pi)^3} \frac{1 - f_{\mathbf{p}} - f_{\mathbf{q}-\mathbf{p}}}{\Omega + i0^+ - \xi_{\mathbf{p}} - \xi_{\mathbf{q}-\mathbf{p}}},$$

with

$$\phi_0''(\mathbf{q}, \Omega) = -\pi \int_{\text{BZ}} \frac{d^3 p}{(2\pi)^3} [1 - f_{\mathbf{p}} - f_{\mathbf{q}-\mathbf{p}}] \delta(\Omega - \xi_{\mathbf{p}} - \xi_{\mathbf{q}-\mathbf{p}}).$$

Here

$$f_{\mathbf{k}} = f(\xi_{\mathbf{k}}) = \frac{1}{e^{\beta \xi_{\mathbf{k}}} + 1}, \quad n_B(\Omega) = \frac{1}{e^{\beta \Omega} - 1}.$$

For powers of the bubbles define

$$X_n''(\mathbf{q}, \Omega) = \text{Im} [\chi_0^R(\mathbf{q}, \Omega)]^n,$$

$$P_n''(\mathbf{q}, \Omega) = \text{Im} [\phi_0^R(\mathbf{q}, \Omega)]^n.$$

Thus

$$X_1'' = \chi_0'',$$

$$X_2'' = 2\chi_0' \chi_0'',$$

$$X_3'' = 3(\chi_0')^2 \chi_0'' - (\chi_0'')^3,$$

and analogously for P_2'', P_3'' .

2. Second order on the real axis

The retarded second-order self-energy is

$$\Sigma^{(2)R}(\mathbf{k}, \omega) = U^2 \int_{\text{BZ}} \frac{d^3 q}{(2\pi)^3} \int_{-\infty}^{\infty} \frac{d\Omega}{\pi} \frac{\chi_0''(\mathbf{q}, \Omega) [n_B(\Omega) + 1 - f_{\mathbf{k}-\mathbf{q}}]}{\omega + i0^+ - \Omega - \xi_{\mathbf{k}-\mathbf{q}}}$$

This is equivalent to the fully energy-resolved expression

$$\Sigma^{(2)R}(\mathbf{k}, \omega) = U^2 \int_{\mathbf{p}} \int_{\mathbf{q}} \frac{f_{\mathbf{p}}(1 - f_{\mathbf{p}+\mathbf{q}})(1 - f_{\mathbf{k}-\mathbf{q}}) + (1 - f_{\mathbf{p}})f_{\mathbf{p}+\mathbf{q}}f_{\mathbf{k}-\mathbf{q}}}{\omega + i0^+ + \xi_{\mathbf{p}} - \xi_{\mathbf{p}+\mathbf{q}} - \xi_{\mathbf{k}-\mathbf{q}}}$$

where

$$\int_{\mathbf{p}} \equiv \int_{\text{BZ}} \frac{d^3 p}{(2\pi)^3}.$$

Therefore

$$\text{Im } \Sigma^{(2)R}(\mathbf{k}, \omega) = -\pi U^2 \int_{\mathbf{p}} \int_{\mathbf{q}} F^{(2)} \delta(\omega + \xi_{\mathbf{p}} - \xi_{\mathbf{p}+\mathbf{q}} - \xi_{\mathbf{k}-\mathbf{q}}),$$

with

$$F^{(2)} = f_{\mathbf{p}}(1 - f_{\mathbf{p}+\mathbf{q}})(1 - f_{\mathbf{k}-\mathbf{q}}) + (1 - f_{\mathbf{p}})f_{\mathbf{p}+\mathbf{q}}f_{\mathbf{k}-\mathbf{q}}.$$

The real part is the principal value of the same denominator.

3. RPA-type particle-hole diagrams on the real axis

The diagrams

$$(2), \quad (3\text{RPA}), \quad (4A) + (4B)$$

combine compactly as

$$\Sigma_{\chi}^R(\mathbf{k}, \omega) = \int_{\mathbf{q}} \int_{-\infty}^{\infty} \frac{d\Omega}{\pi} \frac{[U^2 X_1'' + U^3 X_2'' + 2U^4 X_3''](\mathbf{q}, \Omega) [n_B(\Omega) + 1 - f_{\mathbf{k}-\mathbf{q}}]}{\omega + i0^+ - \Omega - \xi_{\mathbf{k}-\mathbf{q}}}$$

where

$$X_n''(\mathbf{q}, \Omega) = \text{Im} [\chi_0^R(\mathbf{q}, \Omega)]^n.$$

The factor 2 in the U^4 term comes from the two equivalent fourth-order particle-hole RPA diagrams (4A) and (4B).

4. Particle-particle ladder / vertex-channel diagrams

For the particle-particle channel,

$$(3VC), \quad (4C),$$

the internal fermion line is $G_0(q - k)$, so the analytic continuation has a different denominator:

$$\Sigma_{\phi}^R(\mathbf{k}, \omega) = \int_{\mathbf{q}} \int_{-\infty}^{\infty} \frac{d\Omega}{\pi} \frac{[U^3 P_2'' + U^4 P_3''](\mathbf{q}, \Omega) [n_B(\Omega) + f_{\mathbf{q}-\mathbf{k}}]}{\Omega - \omega - \xi_{\mathbf{q}-\mathbf{k}} - i0^+}$$

where

$$P_n''(\mathbf{q}, \Omega) = \text{Im} [\phi_0^R(\mathbf{q}, \Omega)]^n.$$

This is the real-axis version of

$$\int_q \phi_0(q)^2 G_0(q - k), \quad \int_q \phi_0(q)^3 G_0(q - k).$$

5. Fourth-order self-energy insertion diagrams (4D), (4E), (4F)

Define the U -stripped second-order retarded self-energy

$$\sigma_2^R(\mathbf{p}, \epsilon) = \frac{1}{U^2} \Sigma^{(2)R}(\mathbf{p}, \epsilon),$$

and the composite spectral density

$$A_{M_2}(\mathbf{p}, \epsilon) = -\frac{1}{\pi} \text{Im} \left\{ [G_0^R(\mathbf{p}, \epsilon)]^2 \sigma_2^R(\mathbf{p}, \epsilon) \right\}.$$

Then

$$G_0^R(\mathbf{p}, \epsilon) = \frac{1}{\epsilon + i0^+ - \xi_{\mathbf{p}}}.$$

The three insertion diagrams are

$$\Sigma_{4D}^R(\mathbf{k}, \omega) = U^4 \int_{\mathbf{p}} \int \frac{d\Omega}{\pi} \int d\epsilon \frac{\chi_0''(\mathbf{k} - \mathbf{p}, \Omega) A_{M_2}(\mathbf{p}, \epsilon) [n_B(\Omega) + 1 - f(\epsilon)]}{\omega + i0^+ - \Omega - \epsilon}$$

$$\Sigma_{4E}^R(\mathbf{k}, \omega) = U^4 \int_{\mathbf{p}} \int \frac{d\Omega}{\pi} \int d\epsilon \frac{\chi_0''(\mathbf{p} - \mathbf{k}, \Omega) A_{M_2}(\mathbf{p}, \epsilon) [n_B(\Omega) + f(\epsilon)]}{\omega + i0^+ + \Omega - \epsilon}$$

$$\Sigma_{4F}^R(\mathbf{k}, \omega) = U^4 \int_{\mathbf{p}} \int \frac{d\Omega}{\pi} \int d\epsilon \frac{\phi_0''(\mathbf{k} + \mathbf{p}, \Omega) A_{M_2}(\mathbf{p}, \epsilon) [n_B(\Omega) + f(\epsilon)]}{\Omega - \omega - \epsilon - i0^+}$$

In practice these are numerically delicate because G_0^{R2} gives derivative-like spectral weight. If you use a finite broadening η ,

$$G_0^R(\epsilon) = \frac{1}{\epsilon + i\eta - \xi_{\mathbf{p}}},$$

then A_{M_2} is regularized automatically.

6. Fourth-order diagrams $(4G)$, $(4H)$, $(4I)$, $(4J)$, $(4K)$, $(4L)$

For these diagrams the real-axis forms are exact but not compact if fully expanded into thermal factors. The clean way is to use spectral densities and residue kernels.

Define the standard bosonic spectral densities

$$\rho_\chi(\mathbf{q}, \Omega) = -\frac{1}{\pi} \chi_0''(\mathbf{q}, \Omega), \quad \rho_\phi(\mathbf{q}, \Omega) = -\frac{1}{\pi} \phi_0''(\mathbf{q}, \Omega).$$

Also define the residue operators

$$\mathcal{R}_B^z F(z) = - \sum_{z_i} n_B(z_i) \text{Res}_{z=z_i} F(z),$$

$$\mathcal{R}_F^z F(z) = \sum_{z_i} f(z_i) \text{Res}_{z=z_i} F(z),$$

where z_i are the poles of F , not Matsubara-grid poles. After applying these residue operations, set

$$i\omega_n \rightarrow \omega + i0^+.$$

This gives an exact real-axis expression without infinite Matsubara sums.

Diagram $(4G)$

Let

$$\epsilon_1 = \xi_{\mathbf{k}-\mathbf{q}_1}, \quad \epsilon_2 = \xi_{\mathbf{k}-\mathbf{q}_2}, \quad \epsilon_3 = \xi_{\mathbf{k}-\mathbf{q}_1-\mathbf{q}_2}.$$

Then

$$\Sigma_{4G}^R(\mathbf{k}, \omega) = U^4 \int_{\mathbf{q}_1} \int_{\mathbf{q}_2} \int d\Omega_1 \int d\Omega_2 \rho_\chi(\mathbf{q}_1, \Omega_1) \rho_\chi(\mathbf{q}_2, \Omega_2) K_G^R,$$

with

$$K_G^R = \mathcal{R}_B^{z_1} \mathcal{R}_B^{z_2} \left[\frac{1}{(z_1 - \Omega_1)(z_2 - \Omega_2)(i\omega_n - z_1 - \epsilon_1)(i\omega_n - z_2 - \epsilon_2)(i\omega_n - z_1 - z_2 - i\omega_n)} \right]$$

Diagrams $(4H) + (4I)$

Let

$$\epsilon_1 = \xi_{\mathbf{k}-\mathbf{q}_1}, \quad \epsilon_2 = \xi_{\mathbf{q}_2-\mathbf{k}}, \quad \epsilon_3 = \xi_{\mathbf{q}_1+\mathbf{q}_2-\mathbf{k}}.$$

Then

$$\Sigma_{4H}^R + \Sigma_{4I}^R = -2U^4 \int_{\mathbf{q}_1} \int_{\mathbf{q}_2} \int d\Omega_1 \int d\Omega_2 \rho_\chi(\mathbf{q}_1, \Omega_1) \rho_\phi(\mathbf{q}_2, \Omega_2) K_{HI}^R,$$

where

$$K_{HI}^R = \mathcal{R}_B^{z_1} \mathcal{R}_B^{z_2} \left[\frac{1}{(z_1 - \Omega_1)(z_2 - \Omega_2)(i\omega_n - z_1 - \epsilon_1)(z_2 - i\omega_n - \epsilon_2)(z_1 + z_2 - i\omega_n)} \right]$$

Diagram $(4J)$

Let

$$\epsilon_1 = \xi_{\mathbf{p}_1}, \quad \epsilon_2 = \xi_{\mathbf{p}_2}, \quad \epsilon_3 = \xi_{\mathbf{q}-\mathbf{p}_1}, \quad \epsilon_4 = \xi_{\mathbf{q}-\mathbf{p}_2}, \quad \epsilon_5 = \xi_{\mathbf{q}-\mathbf{k}}.$$

Then

$$\Sigma_{4J}^R(\mathbf{k}, \omega) = U^4 \int_{\mathbf{q}} \int_{\mathbf{p}_1} \int_{\mathbf{p}_2} \int d\Omega \rho_\chi(\mathbf{q} - \mathbf{p}_1 - \mathbf{p}_2, \Omega) K_J^R,$$

with

$$K_J^R = \mathcal{R}_B^z \mathcal{R}_F^u \mathcal{R}_F^v \left[\frac{1}{(z - u - v - \Omega)(u - \epsilon_1)(v - \epsilon_2)(z - u - \epsilon_3)(z - v - \epsilon_4)(z - u - v - \epsilon_5)} \right]$$

Diagram $(4K)$

Let

$$\epsilon_1 = \xi_{\mathbf{p}_1}, \quad \epsilon_2 = \xi_{\mathbf{p}_2}, \quad \epsilon_3 = \xi_{\mathbf{q}+\mathbf{p}_1}, \quad \epsilon_4 = \xi_{\mathbf{q}+\mathbf{p}_2}, \quad \epsilon_5 = \xi_{\mathbf{k}-\mathbf{q}}.$$

Then

$$\Sigma_{4K}^R(\mathbf{k}, \omega) = U^4 \int_{\mathbf{q}} \int_{\mathbf{p}_1} \int_{\mathbf{p}_2} \int d\Omega \rho_\phi(\mathbf{q} + \mathbf{p}_1 + \mathbf{p}_2, \Omega) K_K^R,$$

where

$$K_K^R = \mathcal{R}_B^z \mathcal{R}_F^u \mathcal{R}_F^v \left[\frac{1}{(z + u + v - \Omega)(u - \epsilon_1)(v - \epsilon_2)(z + u - \epsilon_3)(z + v - \epsilon_4)(i\omega_r - z)} \right]$$

Diagram (4L)

Let

$$\epsilon_1 = \xi_{\mathbf{p}_1}, \quad \epsilon_2 = \xi_{\mathbf{p}_2}, \quad \epsilon_3 = \xi_{\mathbf{q}+\mathbf{p}_1}, \quad \epsilon_4 = \xi_{\mathbf{q}+\mathbf{p}_2}, \quad \epsilon_5 = \xi_{\mathbf{k}-\mathbf{q}}.$$

Then

$$\Sigma_{4L}^R(\mathbf{k}, \omega) = -U^4 \int_{\mathbf{q}} \int_{\mathbf{p}_1} \int_{\mathbf{p}_2} \int d\Omega \rho_\chi(\mathbf{p}_1 - \mathbf{p}_2, \Omega) K_L^R,$$

with

$$K_L^R = \mathcal{R}_B^z \mathcal{R}_F^u \mathcal{R}_F^v \left[\frac{1}{(u - v - \Omega)(u - \epsilon_1)(v - \epsilon_2)(z + u - \epsilon_3)(z + v - \epsilon_4)(i\omega_n - z)} \right]$$

These residue-kernel forms are usually the safest exact representation for the non-RPA fourth-order diagrams. Expanding them by hand gives many terms with different Fermi/Bose occupation factors and principal-value denominators.

7. Real and imaginary parts

For any term written as

$$\int d\Omega \frac{F(\Omega)}{\omega + i0^+ - \Omega - \epsilon},$$

one has

$$\text{Im} = -\pi F(\omega - \epsilon),$$

and

$$\text{Re} = \mathcal{P} \int d\Omega \frac{F(\Omega)}{\omega - \Omega - \epsilon}.$$

For terms with denominator

$$\Omega - \omega - \epsilon - i0^+,$$

one instead has

$$\text{Im} = +\pi F(\omega + \epsilon),$$

because

$$\frac{1}{x - i0^+} = \mathcal{P} \frac{1}{x} + i\pi\delta(x).$$

Numerically, it is often best to compute

$$\text{Im} \Sigma^R(\mathbf{k}, \omega)$$

from the delta-function forms, then obtain

$$\text{Re} \Sigma^R(\mathbf{k}, \omega)$$

by a Kramers-Kronig transform, adding the Hartree constant separately:

$$\Sigma_{\text{Hartree}} = U n_{-\sigma}.$$

8. Practical numerical prescription

For a direct real-axis calculation:

1. Precompute

$$\chi_0^R(\mathbf{q}, \Omega), \quad \phi_0^R(\mathbf{q}, \Omega)$$

on a grid of $\mathbf{q}$ and real Ω .

2. Store both real and imaginary parts. You need $\chi', \chi'', \phi', \phi''$, because fourth-order terms contain

$$\text{Im}[\chi^R]^2, \quad \text{Im}[\chi^R]^3, \quad \text{Im}[\phi^R]^2, \quad \text{Im}[\phi^R]^3.$$

3. Use the compact one-boson-convolution formulas for

$$(2), \quad (3\text{RPA}), \quad (3\text{VC}), \quad (4A), (4B), (4C).$$

4. Treat $(4D), (4E), (4F)$ using the spectral function of

$$[G_0^R]^2 \sigma_2^R.$$

5. Treat $(4G)-(4L)$ either by the residue kernels above or by evaluating them on the Matsubara axis and analytically continuing the final result.

The main warning is that the real-axis fourth-order vertex/envelope diagrams contain many principal-value structures. They are exact, but substantially less convenient than the Matsubara expressions unless you only need $\text{Im } \Sigma^R$.